ModelIt!

Teacher's Guide

# The Opioid Epidemic: When Medicine Becomes the Disease

Modeling Neurochemical Feedback Loops in Addiction

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 09L2 |
| **NGSS Standard** | HS-LS1-2, HS-ETS1-4 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |


© 2026 Discovery Collective. All rights reserved. For teacher use only.

# Table of Contents

ModelIt!


1 NGSS Standards Unpacking & CAST Alignment ..... 3

2 Scoring Rubric ..... 4

3 Background Content Knowledge ..... 5

4 The LEVER Framework ..... 6

5 Slide-by-Slide Facilitation Guide ..... 7

6 Answer Key & Expected Responses ..... 9

7 Collaborative Learning Protocols ..... 10

8 Differentiation & Extensions ..... 11

9 Common Misconceptions ..... 12

# 1 NGSS Standards & CAST Alignment

ModelIt!

**Performance Expectation: HS-LS1-2, HS-ETS1-4**

Develop and use a model to illustrate the hierarchical organization of interacting systems. Use a computer simulation to model the impact of proposed solutions to a complex real-world problem.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Developing and Using Models** — Students build multi-feedback-loop models of neurochemical tolerance, dependence, and social network epidemic effects. |
| **Disciplinary Core Idea** | **LS1.A Structure and Function** — The brain dopamine reward system adapts to chronic opioid exposure through receptor downregulation. |
| **Crosscutting Concept** | **Cause and Effect: Mechanism and Prediction** — Students trace the causal chain from prescription to dependence to epidemic, identifying threshold effects. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

Which feedback loop is MOST responsible for driving dose escalation?

**CONSTRUCTED RESPONSE:**

Explain why a patient on opioids for 6 months faces a fundamentally different neurochemical situation than one on them for 2 weeks.

# 2 Scoring Rubric

ModelIt!

### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

ModelIt!

**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The opioid epidemic has killed over 500,000 Americans since 1999. Understanding how prescribed medicine becomes lethal addiction requires modeling neurochemical feedback loops that hijack the brain reward system.

## The Neurochemistry of Tolerance

Opioids bind to mu-opioid receptors, triggering dopamine release. With repeated exposure, the brain reduces receptor number and sensitivity. This is tolerance: the same dose produces less effect.

## The Dependence Threshold

After sufficient exposure, the brain baseline neurochemistry shifts. It now REQUIRES opioids to function normally. Without them, severe withdrawal occurs.

## Opioid-Induced Hyperalgesia

Chronic opioid use can paradoxically INCREASE pain sensitivity. The brain compensates by amplifying pain signals, creating a second feedback loop.

## The Social Network Effect

Leftover pills became community exposure vectors. When prescriptions tightened, demand shifted to illicit heroin and fentanyl.

# 4 The LEVER Framework

ModelIt!

### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** — **Locate the System**
Students identify the 7 interacting components of the opioid crisis system.

**E** — **Establish Relationships**
Students map multiple feedback loops: tolerance-dose escalation, hyperalgesia-pain-use, and social exposure-new users.

**V** — **Visualize & Model**
Students build a computational model with 7 components, identifying reinforcing loops and thresholds.

**E** — **Evaluate Outcomes**
Students run short-term, chronic, and community-spread scenarios to test intervention timing.

**R** — **Revise & Extend**
Students add naloxone, MAT, or fentanyl contamination to model real-world interventions.

## Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

ModelIt!

## Slide 1: Cover

2 min

**Say:** *"Every day, 130 Americans die from opioid overdose. Most started with a doctor prescription."*

**Watch For**

Show the statistic. Establish safe, scientific frame.

## Slide 2: Learning Objectives

3 min

**Say:** *"Today we model what happens in the brain with repeated opioid exposure. Neuroscience, not moral judgment."*

**Watch For**

Pre-teach tolerance, dependence, receptor downregulation.

## Slide 3: Big Question

3 min

**Say:** *"Doctors prescribed these drugs to help. How did helping become harming?"*

**Watch For**

Quick-write: What do you know about the opioid crisis?

## Slide 4: LEVER Framework

2 min

**Say:** *"This model has MULTIPLE feedback loops. That makes it Level 2."*

**Watch For**

Preview: 7 components, multiple reinforcing loops, threshold effects.

## Slide 5: Activity 1: Components

8 min

**Say:** *"Which can doctors or policymakers control? Which are internal brain processes?"*

**Watch For**

Guide sorting of external vs internal.

## Slide 6: Activity 2: Connect

10 min

**Say:** *"This system has at least THREE reinforcing feedback loops. Find them."*

**Watch For**

Identify tolerance loop, hyperalgesia loop, social spread loop.

## Slide 7: Activity 3: Simulate

10 min

**Say:** *"Find the dependence threshold. When does the brain cross the point of no return?"*

**Watch For**

Students run scenarios and identify threshold transitions.

## Slide 8: Discoveries

5 min

**Say:** *"Your model shows why addiction is a disease, not a choice."*

**Watch For**

Evidence-based discussion connecting model to real data.

## Slide 9: STEM Challenge

12 min

**Say:** *"You have $50 million for a state in crisis. Where do you invest?"*

**Watch For**

Groups design dashboards. Justify with model evidence.

# 6 Answer Key & Expected Responses

ModelIt!

## Component Sorting

- **EXTERNAL:** Prescription Rate, Social Network Exposure
- **INTERNAL:** Tolerance Development, Addiction Level, Dopamine Receptor Sensitivity, Pain Threshold, Withdrawal Severity

## Relationships

- **Prescription Rate to Tolerance:** POSITIVE (+) — Higher prescriptions mean more exposure, accelerating tolerance.
- **Tolerance to Addiction Level:** POSITIVE (+) — Higher tolerance requires higher doses, deepening dependence.
- **Addiction to Dopamine Sensitivity:** NEGATIVE (-) — Increasing addiction drives receptor downregulation.
- **Dopamine Sensitivity to Tolerance:** NEGATIVE (-) — Less sensitive receptors mean same dose does less, driving tolerance up.
- **Addiction to Pain Threshold:** NEGATIVE (-) — Chronic use causes hyperalgesia, lowering pain threshold.
- **Pain Threshold to Prescription Rate:** NEGATIVE (-) — Lower pain thresholds drive patients to seek more medication.
- **Social Exposure to Addiction:** POSITIVE (+) — Greater community exposure increases non-medical use.
- **Addiction to Withdrawal Severity:** POSITIVE (+) — Deeper addiction produces more severe withdrawal.

## Key Discoveries

- Opioid tolerance creates a reinforcing feedback loop: reduced sensitivity requires higher dose which causes faster adaptation
- There is a neurochemical threshold past which dependence becomes physiological
- Chronic opioid use paradoxically INCREASES pain sensitivity via hyperalgesia
- Social network effects amplify a medical problem into an epidemic

# 7 Collaborative Learning Protocols

ModelIt!

### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

ModelIt!

### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

ModelIt!

**Why This Matters**

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Addiction is a moral failure" | The feedback model shows addiction is a predictable neurochemical outcome. Past the dependence threshold, willpower cannot override altered neurochemistry. |
| "Only certain people become addicted" | Anyone with a brain can develop dependence. The variables are dosage, duration, and genetics, not character. |
| "People take opioids to get high" | Past the dependence threshold, people take opioids to avoid feeling terrible, not to feel good. Motivation shifts from reward-seeking to punishment-avoidance. |